

Multi-layer Perceptrons (MLP) with BP Training

Exercise 1.

The example program (demo) demonstrates the capability of an MLP to model the XOR logic gate. Open the demo.m program and then go through it. There are 4 clusters of data (A,B,C,D) are defined in a 2-dimensional input space. (A,C) and (B,D) clusters represent XOR classification problem. Understand the codes and experiment with different configurations.

Exercise 2.

In this exercise, you are required to demonstrate the capability of an MLP to approximate the function

$$f(t) = \sin(t) \cdot \exp(-t/20); 0 < t < 50$$

Experiment with different number of training data points, different number of hidden layers and nodes, etc.

Task: MLP model for the printed digits classification

You are given a set of printed digits (0-5) (training and testing). Each digit is in binary format. After using imread to load an image into memory (e.g., `im=imread('0a.bmp')`), you can display the image using imshow command. Then the image matrix can be changed into a vector by simply using `x=im(:)`.

After training using MLP network, the trained model should be simulated using the function `sim`:

```
>> y = sim(net, x); % where x is the testing data
```

The measurement of the rate of correct classified vectors can be done as follows based on the provided `getcls` function:

```
>> rate(getcls(y), t);
```

Try to classify all training vectors (using `sim`) and record the classification result (using `getcls` and `rate`) with different number of layers, neurons and epochs, based on the three provided m-files: `getimdata.m`, `getcls.m` and `rate.m`. One possible solution is provided for your reference: `main.m`.